are costly, slow, or hazardous. As urban populations expand and cities pursue smarter, more efficient service models, drones are emerging as essential tools for creating resilient, responsive, and technologically advanced urban environments.

8.7.1 Aerial Inspectors

Urban infrastructure, such as bridges, flyovers, railway tracks, pipelines, cellular towers, and power lines, requires regular inspection to ensure structural safety and operational continuity. Traditionally, these inspections rely on scaffolding, cranes, manual climbing, or temporary shutdown methods that are labor-intensive, time-consuming, and sometimes dangerous. Drones equipped with high-resolution imaging, thermal sensors, and LiDAR mapping systems allow engineers to conduct detailed structural assessments quickly and without service interruption.

These aerial inspections detect micro-cracks, corrosion, misalignments, insulation faults, and heat leakage patterns, enabling proactive maintenance before failures occur. Moreover, repeated drone surveys create digital time-series records that support predictive maintenance models and digital twin infrastructure simulations. By minimizing inspection time, reducing worker exposure to hazards, and lowering maintenance costs, UAV-based monitoring enhances the reliability and sustainability of critical urban infrastructure networks.

8.7.2 Integrating UAVs into smart cities

Drones play an important role in improving urban traffic monitoring and mobility planning. Real-time aerial imagery enables traffic control centers to track vehicle flow patterns, identify congestion hotspots, monitor accidents, and evaluate the effectiveness of traffic signal timings. This